Robust Open-Set CAN Attack Classification under Domain Shift: A Dual-Alignment Framework

***Abstract*—To address the challenges of domain shift across in-vehicle Controller Area Network (CAN) and the difficulty of identifying unknown attack types in open-set scenarios, this paper proposes a cross-domain intrusion detection network (DORU-Net) based on dual alignment and open-set rejection. The method first reconstructs CAN messages using grayscale transformation and extracts high-dimensional spatiotemporal features through a convolutional neural network. In the feature alignment stage, an architecture with a Domain Adversarial Neural Network (DANN) as its backbone is constructed, combined with a Gradient Reversal Layer (GRL) and CORAL loss, achieving dual constraints on the source and target domains in both second-order statistical distribution and adversarial feature space, significantly enhancing the robustness of alignment. In the inference stage, a cascaded mechanism of "known class determination—OpenMax adaptive rejection" is designed to address the risks in non-stationary open spaces. Regardless of whether the target domain faces a single unknown attack or a mixture of unknown attacks, it can uniformly identify them as "unknown classes" through distribution boundaries. This paper proposes a robust framework for security monitoring, targeting the pervasive non-independent and identically distributed (Non-IID) data characteristics in connected vehicle systems.**

***Index Terms*—CAN, Intrusion Detection, Domain Adaptation, Open-Set Recognition**

I. Introduction

With the advancement of Intelligent Connected Vehicles (ICVs) and autonomous driving technologies, each vehicle has evolved as a cyber-physical node, and the entire vehicular ecosystem has further formed a large-scale interconnected Cyber-Physical System. As the backbone of in-vehicle communication, the Controller Area Network (CAN) is responsible for transmitting critical commands such as powertrain, chassis, and body control [1]. However, the CAN protocol lacked encryption and authentication mechanisms at its inception, making it highly vulnerable to malicious attacks such as spoofed message injection and replay, which pose serious threats to driving safety [2].

Although deep learning-based Intrusion Detection Systems (IDS) have achieved remarkable performance in attack identification for dedicated vehicle models [3], their cross-vehicle generalization is severely constrained by domain heterogeneity. For CAN intrusion detection, **cross-domain** characterizes the distinct data distributions derived from disparate vehicle models, signal coding specifications, CAN ID allocation schemes and driving scenarios. We define the **source domain** as the labeled CAN dataset (including normal and attack samples) used for model training, and the **target domain** as the unseen vehicle platforms and practical driving environments where the detection model is deployed. Against this background, two critical bottlenecks remain unresolved in real-world applications:

1) Domain Shift caused by Non-IID Data: CAN messages under different vehicle models or operating conditions exhibit significant differences in signal encoding and ID distribution. When models trained in the source domain are directly transferred to the target domain, performance often degrades drastically due to mismatched feature distributions [4].
2) Open-Set Recognition Problem: Existing methods mostly rely on the closed-set assumption, where training and testing sets share the same label space [5]. However, in real-world scenarios, the target domain often encounters unknown attacks never seen in the source domain. Traditional classifiers tend to force these unknown samples into known categories, leading to high-risk misclassifications.

Therefore, constructing a highly robust intrusion detection model in complex scenarios where cross-vehicle domain migration coexists with open-set unknown attacks is a critical issue that urgently needs to be addressed in automotive cybersecurity.

Aimed at the two critical practical bottlenecks in real deployment—cross-domain distribution shift induced by Non-IID CAN data and open-set recognition with unknown attacks—existing dedicated studies can be classified into three groups with distinct but complementary limitations:

- Statistical Feature-based Approaches: Relying on handcrafted features such as message interval and ID frequency [6], these methods are designed for static single-domain closed-set scenarios. They possess no cross-domain generalization ability against Non-IID distribution discrepancies, and lack any mechanism for unknown attack detection, thus being unfit for the coupled challenge scenario.
- Closed-Set Deep Learning Approaches: Spatiotemporal feature extraction models (CNNs, LSTMs, etc.) achieve superior performance on homogeneous in-domain datasets [7], yet they neglect domain shift across different vehicle models and operating conditions. Moreover, constrained by the closed-set hypothesis, they cannot distinguish novel unknown attacks, resulting in severe performance degradation under the joint influence of cross-domain and open-set variations.
- Closed-Set Domain Adaptation Approaches: Adversarial training-based domain alignment is adopted to mitigate feature distribution mismatch between source and target

domains [8], which relieves the domain shift problem caused by Non-IID CAN messages. However, these methods are inherently bounded by the shared label space assumption. The emergence of unknown attack classes in the target domain will induce severe *Negative Transfer* and *Cluster Collapse*, meaning they only resolve the cross-domain challenge while completely failing the open-set recognition task.

Addressing the aforementioned challenges, this paper proposes a Domain-Adaptive Open-Set Rejection-based Unified Cross-Domain Intrusion Detection Network (DORU-Net). This framework aims to achieve precise migration of known attacks while effectively intercepting unknown threats without requiring labels from the target domain. The main contributions of this paper are as follows:

1) **Construction of a Dual Feature Alignment Architecture:** We propose an adversarial adaptation network combining a Gradient Reversal Layer (GRL) with CORAL loss [9]. The GRL achieves first-order distribution alignment of the feature space, while CORAL constrains second-order statistics (covariance) [10]. Their synergistic operation significantly enhances the feature invariance and robustness of the model in cross-vehicle scenarios.
2) **Design of a Cascaded Open-Set Inference Mechanism:** We innovatively design an inference strategy comprising "Known Class Determination" followed by "OpenMax Adaptive Rejection." Utilizing Extreme Value Theory (EVT) to fit the tail distribution of activation vectors, we calculate the probability of a sample belonging to an unknown class [6]. This mechanism ensures recognition accuracy for known attacks while flexibly intercepting unseen attacks as generic unknown threats.

The rest of the paper is organized as follows. Section II elaborates on the preliminary work. Section III details the proposed model architecture. Section IV presents and discusses the experimental result evaluation. Finally, Section V summarizes the whole paper and puts forward future research directions.

II. Preliminaries

This section presents the fundamental theories and key technologies underpinning DORU-Net, focusing on four core aspects to support subsequent model design and experiments. It first elaborates on the CAN protocol, its message structure, common attack types, and utilized datasets, providing a practical background for in-vehicle network intrusion detection. Then, it explains domain adaptation, the DANN framework, and the role of GRL in addressing cross-vehicle domain shift. Next, the CORAL method is introduced to complement feature distribution statistical alignment. Finally, it covers open-set recognition and the OpenMax algorithm, laying the theoretical foundation for the model to handle unseen threats.

A. Controller Area Network (CAN)

1) Message Structure: Controller Area Network [11] is a broadcast-oriented, multi-master bus communication protocol widely used for data exchange between automotive electronic control units (ECUs). The key fields of a standard CAN message are shown in Table I.

TABLE I: Frame Structure of CAN Message

Start of Frame	Arbitration Field	Control Field	Data Field	CRC	ACK	End of Frame
1 Bit	12 Bits	6 Bits	0-64 Bits	16 Bits	2 Bits	7 Bits

2) CAN Attack Types and Threat Model: The specific characteristics and potential consequences of these In-Vehicle Network (IVN) attacks are detailed in the TABLE II below [12] [13].

TABLE II: Description of Attack Types and Normal Traffic

Traffic Type	Description
DoS Attack	Floods the network with high-priority messages to block normal traffic transmission, causing potential loss of control.
Fuzzy Attack	Injects messages with arbitrary IDs and data (sometimes mimicking valid IDs) to disrupt system stability unpredictably.
Gear Spoofing	Floods the bus with gear-control related IDs and random data to mislead the ECU regarding gear operations.
RPM Spoofing	Overrides legitimate signals using tachometer-linked IDs to manipulate displayed RPM readings on the dashboard.
Malfunction Attack	Modifies data fields of specific extracted IDs and injects them to trigger abnormal vehicle behaviors.
Normal Traffic	Standard network traffic generated during the vehicle's regular operation without malicious interference.

In this work, it assumes that the source domain only includes known categories such as normal packets and DoS attacks, while the target domain additionally includes attack types not seen in the source domain, such as fuzzy attacks and malfunction attacks, thus forming an open-set domain adaptation scenario.

B. Domain Adaptation

1) Definition of the domain adaptation problem: Let the source domain be defined as

$$D_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s} \tag{1}$$

,where D_s denotes the labeled source domain dataset, where each element (x_i^s, y_i^s) corresponds to one sample from the source domain: x_i^s is the feature vector of the i-th source sample, y_i^s is its category label, and n_s is the total number of samples in D_s. All these samples are independently and identically drawn from the joint probability distribution $P_s(X, Y)$.

The target domain dataset is defined as

$$D_t = \{x_j^t\}_{j=1}^{n_t} \tag{2}$$

,where D_t denotes the unlabeled target domain dataset, where each x_j^t is the feature vector of the j-th target sample, and n_t is the total number of samples in D_t. These samples are

independently and identically drawn from the joint probability distribution $P_t(X, Y)$, and their category labels are typically unavailable or only sparsely provided.

Due to the discrepancies in CAN frame ID distributions and data characteristics across distinct vehicle models, this commonly leads to

$$P_s(X, Y) \neq P_t(X, Y) \tag{3}$$

, which constitutes the domain shift problem.

Unsupervised domain adaptation (UDA) aims to improve the classification accuracy on the target domain. To achieve this objective, we leverage labeled data from the source domain and unlabeled data from the target domain to learn a feature extractor $F(\cdot)$ and a classifier $C(\cdot)$. A key design principle is to ensure that the features extracted by $F(\cdot)$ exhibit highly aligned distributions across the two domains [14], thereby facilitating the generalization of the classifier $C(\cdot)$ to the target domain.

2) Domain-Adversarial Neural Network: DANN [14] is a typical adversarial domain adaptation framework. The core idea is to simultaneously train two modules on a shared feature extractor F.

1. Source Domain Classifier C performs supervised classification on features from the source domain.

2. Domain Discriminator G determines whether the input features originate from the source domain or the target domain.

Through adversarial training, the objective is for the feature extractor F to learn features that are both effective for source domain task classification and difficult for the domain discriminator G to determine their origin. This enables the learning of domain-invariant representations.

In the DANN framework, the source domain classification loss is defined as

$$L_{\text{cls}} = \frac{1}{n_s} \sum_{i=1}^{n_s} l(C(F(x_i^s)), y_i^s) \tag{4}$$

, where $l(\cdot)$ is typically the cross-entropy loss.

The domain discrimination loss is defined as

$$L_{\text{dom}} = \frac{1}{n_s + n_t} \sum_{k} l(G(F(x_k)), g_k) \tag{5}$$

, where $g_k \in \{0, 1\}$ is the domain label, where $g_k = 0$ denotes the source domain and $g_k = 1$ denotes the target domain.

3) Gradient Reversal Layer: To achieve the adversarial training described above, DANN introduces a Gradient Reversal Layer (GRL) [9]. During forward propagation, GRL acts as an identity mapping

$$\text{GRL}(z) = z. \tag{6}$$

During backward propagation, however, GRL multiplies the gradient passing through it by a negative coefficient $-\lambda$

$$\frac{\partial \text{GRL}}{\partial z} = -\lambda I \tag{7}$$

, where I is the identity matrix.

During joint training, the domain discriminator D minimizes the domain discrimination loss L_{dom}, while the feature extractor F, influenced by the GRL, effectively maximizes L_{dom}, pushing the source and target feature distributions to become indistinguishable.

The overall optimization objective can be expressed as

$$\min_{F,C} \max_{D} \quad L_{\text{cls}}(F, C) - \lambda L_{\text{dom}}(F, D) \tag{8}$$

Building upon this framework, we further introduce the statistical alignment loss CORAL, constructing a dual-alignment training strategy.

C. Correlation Alignment

Correlation Alignment (CORAL) [10] is a method for transfer learning that explicitly matches the second-order statistics of features between the source domain and the target domain. The core idea is to align not only the means of feature distributions in the feature space but also their covariance matrices. This alignment ensures that the two domains feature distributions are consistent not only in position but also in internal correlation structures.

Let the feature matrix of the source domain be $H_s \in \mathbb{R}^{n_s \times m}$, and the feature matrix of the target domain be $H_t \in \mathbb{R}^{n_t \times m}$, where m is the feature dimension. The covariance matrices of the source and target domains can be calculated as

$$\Sigma_s = \frac{1}{n_s - 1} \left(H_s^\top H_s - \frac{1}{n_s} (1^\top H_s)^\top (1^\top H_s) \right) \tag{9}$$

$$\Sigma_t = \frac{1}{n_t - 1} \left(H_t^\top H_t - \frac{1}{n_t} (1^\top H_t)^\top (1^\top H_t) \right) \tag{10}$$

where 1 represents a vector of all ones.

The CORAL loss is defined as the Frobenius norm distance between the two covariance matrices

$$L_{\text{coral}} = \frac{1}{4m^2} \|\Sigma_s - \Sigma_t\|_F^2 \tag{11}$$

By incorporating L_{coral} into the training objective, it provides statistical constraints to further align the feature distributions beyond adversarial alignment. This enhances the stability and effectiveness of domain alignment while mitigating issues such as oscillations or feature collapse that adversarial training alone might cause.

In this work, the CORAL loss is combined with the classification loss and domain adversarial loss in the DANN framework through weighted integration, forming a full dual-alignment training objective.

D. Open-set recognition

1) Open-set Recognition Problem: Standard deep classifiers are usually trained with a closed-set assumption: the classes seen during training are the same as those at test time. When novel or unknown classes show up during testing, Softmax-based classifiers often force these samples into one of the known classes, which can lead to dangerous errors in safety-critical settings [15].

Open-set recognition tackles this gap by training models to both classify known classes correctly and reject samples from unknown classes, assigning them to an unknown category at inference [6]. In the CAN security setting described here, unknown classes can include target-domain-only attacks such as fuzzy attack and certain vendor-specific malfunction behaviors that do not appear in the source-domain training data.

2) OpenMax for open-set rejection: We adopt the OpenMax [16] paradigm as a robust post-processing module to support open-set rejection. The key steps are:

1. Class centroids: For each known class c, a class center μ_c is computed in the model's activation space (in the implementation, pre-ReLU features are used, and logits-based activations are also supported).

2. Tail modelling via EVT: For each class, distances $d_c(x)$ of training samples to μ_c are collected. Using Extreme Value Theory (EVT), the tail of the distance distribution is fitted with a Weibull distribution to obtain per-class tail parameters. In practice, fitting is performed only when the tail exhibits sufficient variability; otherwise, fitting is omitted for that class.

3. Test-time scoring: For a test sample x, compute distances $d_c(x)$ and evaluate the Weibull cumulative distribution function (CDF) to obtain a per-class tail probability $w_c(x)$. The per-sample unknown score is taken as $\max_c w_c(x)$ and class-wise attenuation factors $1 - w_c(x)$ are used to recalibrate the original activations.

4. Recalibration and unknown channel: Recalibrate logits/activations by reducing each known-class activation according to $w_c(x)$ and allocate the attenuated mass to an explicit "unknown" channel. The final probability vector is renormalized so that known-class probabilities sum to $1 - p_{\text{unknown}}$ and the last element equals p_{unknown}.

III. PROPOSED SCHEME

DORU-Net comprises three main functional modules, the overall architecture of which is depicted in Figure 1. First, an input representation module encodes raw CAN frames into fixed-size grayscale images and leverages a convolutional network to extract high-dimensional spatio-temporal features. Second, a dual-alignment training module performs adversarial domain alignment with a gradient reversal layer (GRL) and second-order statistical alignment with CORAL, mitigating source–target domain gaps in both first- and second-order statistics during feature learning. Third, an open-set inference module executes a staged decision pipeline that first conducts known-class acceptance and then implements OpenMax adaptive rejection, thus enabling consistent recognition and reliable interception of both single and mixed unknown attacks within the target domain.

A. Data preprocessing

DORU-Net maps raw CAN frame sequences into fixed-size grayscale images, the workflow of this data conversion process is depicted in Figure 2, to fully leverage the strengths of convolutional neural networks in modeling local patterns and benefiting from pretraining transfer [17]. This design turns local temporal–signal variations such as burst injections and periodic perturbations into two-dimensional local structures that convolutional kernels can capture effectively, while also making it easy to reuse off-the-shelf visual backbones for feature learning.

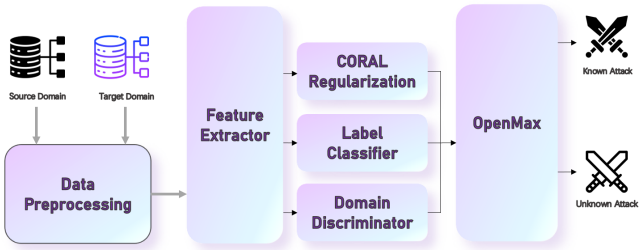

Fig. 1: Frame of DORU-Net

Fig. 2: Data Preprocessing Pipeline

Preprocessing converts raw CAN frames into fixed-size grayscale images. The pipeline comprises three compact steps:

Step 1: Frame standardization and labeling. For every CAN frame we extract and standardize the core fields (`CANID`, `DATA[0]`–`DATA[7]`, and `Label`) so that each record corresponds to one frame with identical field definitions across source and target domains.

Step 2: Normalization and pixel mapping. All numeric fields (`CANID`, `DATA[0]`–`DATA[7]`) are normalized to $[0, 1]$ using the same rule for source and target, then linearly mapped to the grayscale pixel range $[0, 255]$ (0 = black, 255 = white) to produce pixel-compatible values.

Step 3: Tabular-to-image conversion and storage. For each sample, a short window of 9 consecutive frames is stacked (features = `CANID` + `DATA[0]`–`DATA[7]`) to form a 9×9 grayscale matrix, resized to 224×224 via bilinear interpolation for compatibility with standard visual backbones and saved as a PNG, with this compact representation preserving multi-field spatio-temporal anomalies for convolutional feature learning and maintaining consistent conversion rules across domains.

B. DANN-based Domain-Adaptive Feature Learning

This section describes the three core modules for domain-invariant feature learning: the Feature Extractor, the Label Classifier and the Domain Discriminator, whose overall architecture is depicted in Figure 3. These modules are jointly optimized via adversarial training and a statistical alignment

term, yielding learned features that are not only discriminative for the source domain but also robust to domain shifts across cross-vehicle platforms.

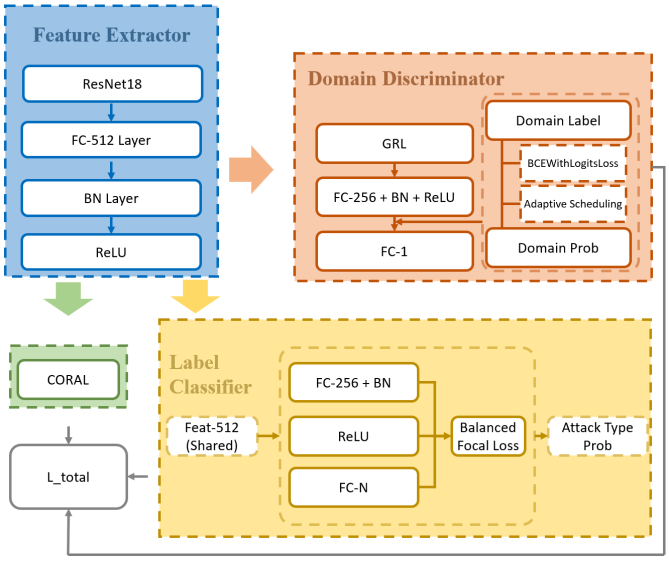

Fig. 3: Frame of DANN

1) Feature extractor: As the foundational module of the DANN framework in Figure 3, the feature extractor is a ResNet-18 backbone without its classification head. Its convolutional output is flattened and projected to a 512-D embedding through a linear layer, BatchNorm1d, and ReLU. Inputs are 224×224 grayscale images replicated to three channels and normalized. For OpenMax fitting and fallback, the extractor can also return pre-ReLU features alongside the usual post-ReLU embeddings. The backbone starts from ImageNet pretrained weights to speed convergence and improve transferability.

2) Label classifier: The label head is a two-layer fully connected network with a 256-unit hidden layer. It consumes the 512-dimensional feature vectors and outputs logits over source classes. Training uses a class-weighted focal loss to mitigate source-domain class imbalance; class weights are computed from source training sample frequencies in the data loader. The classifier and extractor are optimized jointly to maintain discriminability on the source domain.

3) Domain discriminator and GRL: The domain discriminator has a similar two-layer MLP architecture and outputs a single logit for binary domain prediction (source vs. target). A Gradient Reversal Layer (GRL) is inserted between the extractor and the discriminator at the key connection position shown in Figure 3: GRL acts as the identity in the forward pass and multiplies gradients by a negative scalar during backpropagation, thereby encouraging the extractor to produce domain-invariant features. The GRL intensity is scheduled over training progress to first preserve label discriminability and then progressively strengthen domain confusion. The discriminator loss uses BCEWithLogitsLoss. Beyond adversarial alignment, a CORAL term matches the second-order statistics of source and target features, stabilizing alignment and mitigating cluster collapse. The training objective combines class focal loss, adversarial domain loss, and the CORAL term. Separate optimizers handle the feature extractor plus classifier and the discriminator, while cosine-annealing schedulers and early stopping are used to steady convergence.

C. Open-Set Recognition via OpenMax

We integrate OpenMax to reject attack types unseen during training, recalibrating prediction probabilities to account for the chance an input is unknown rather than forcing it into a known class, and the detailed workflow of this OpenMax-based open-set recognition strategy is depicted in Figure 4. After training the feature extractor and classifier on the source

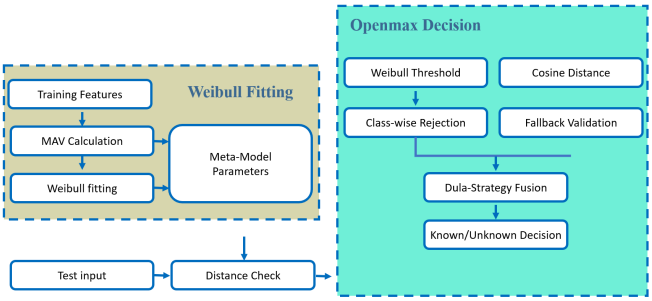

Fig. 4: Frame of Openmax

domain, we compute a centroid for each known class from penultimate-layer features, measure distances from correctly classified samples to their centroid, and fit a Weibull distribution to the tail of these distances to capture the boundary of the known distribution.

At test time, for each incoming CAN packet, the system measures its distances to the class centroids; if it lies far outside all known high-density regions in line with the extreme-value tail, OpenMax shifts probability mass from the known classes to an explicit Unknown class. A sample is flagged as unknown when this calibrated Unknown probability exceeds a preset threshold, giving a unified strategy that detects any attack pattern whose features deviate sufficiently from the learned distributions of known classes.

IV. Experimental Evaluation

This section systematically evaluates DORU-Net with three goals: quantify cross-brand/domain transfer on known attack classes, assess open-set detection for target-domain-only unknown attacks, and test robustness to class imbalance. To ensure reproducibility and statistical reliability, all results are reported as the mean ± standard deviation across multiple random seeds.

A. Experimental Setup

The computing environment and the main training hyperparameters used in all experiments are summarized as follows: the computing environment consists of a Windows 11 (64-bit) operating system, an AMD Ryzen 9 7945HX processor,

an NVIDIA GeForce RTX 4080 GPU, the PyTorch 1.18.0 framework, and Python 3.9 as the programming language. Table III lists the key training hyperparameters and selected implementation notes.

TABLE III: Key training hyperparameters and selected implementation notes

Input size	224×224
Batch size	64
Epochs	50
LR (Feature extractor + C)	1×10^{-3}
LR (Domain discriminator)	1×10^{-4}
Weight decay	1×10^{-4}
Domain loss weight w_{da}	0.5
CORAL weight λ_{coral}	0.1
Focal loss gamma	2.0
OpenMax tailsize	80

The evaluation metrics include Accuracy, Precision, Recall and F1-score, calculated based on the confusion matrix. The specific calculation formulas of each metric are:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{12}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{13}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{14}$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{15}$$

, where TP denotes the number of attack samples correctly classified, TN represents the number of normal samples correctly classified, FP refers to the number of normal samples misclassified as attacks, and FN is the number of attack samples misclassified as normal.

B. Closed-set Evaluation

We first evaluate DORU-Net on closed-set detection using source-domain data (Car-Hacking dataset, Hyundai Sonata split). This measures how well the model separates known traffic categories (Normal R, DoS, RPM spoofing, Gear spoofing) when no target-private classes are present. TableIV reports Accuracy, Precision, Recall, and F1.

As shown in TableIV, DORU-Net delivers near-perfect discrimination on all four categories and outperforms other comparative models, including DCNN [18], HyDL IDS [19], and SupCon ResNet [20], across these attack types. With closed-set performance near ceiling, we next examine cross-domain transfer and open-set detection to assess generalization and robustness to target-private attacks.

C. Open-set Evaluation

For the open-set evaluation, we leveraged two real-world automotive network datasets—Car-Hacking [12] and Survival [13]—chosen for their practical relevance and widespread adoption in related research. The Car-Hacking dataset, serving as our primary source domain data, contains CAN bus traffic from a Hyundai Sonata, encompassing both normal operations and diverse malicious intrusions including Denial-of-Service (DoS), fuzzy, gear spoofing, and RPM spoofing attacks. To simulate a data-scarce target domain scenario, we utilized the Survival dataset, which provides traffic samples from three distinct vehicle models: Hyundai Sonata, Kia Soul, and Chevrolet Spark, with records of normal activity alongside DoS, fuzzy, and malfunction attacks.

TABLE IV: Closed-set malicious traffic detection results on source-domain data

Category	Model	Accuracy	Precision	Recall	F1-score
DoS	DCNN [18]	0.9997	0.9997	0.9997	0.9997
	HyDL IDS [19]	1.0000	1.0000	1.0000	1.0000
	SupCon ResNet [20]	0.9998	0.9998	0.9999	0.9998
	DORU-Net	1.0000	1.0000	1.0000	1.0000
Gear	DCNN [18]	0.9996	0.9994	0.9963	0.9980
	HyDL IDS [19]	1.0000	1.0000	1.0000	1.0000
	SupCon ResNet [20]	0.9994	0.9999	0.9994	0.9997
	DORU-Net	0.9999	0.9999	0.9999	0.9999
RPM	DCNN [18]	0.9995	0.9999	0.9994	0.9996
	HyDL IDS [19]	0.9998	0.9998	0.9999	0.9998
	SupCon ResNet [20]	0.9995	0.9999	0.9994	0.9997
	DORU-Net	1.0000	1.0000	1.0000	1.0000
Fuzzy	DCNN [18]	0.9930	0.9937	0.9984	0.9961
	HyDL IDS [19]	0.9997	0.9998	0.9998	0.9998
	SupCon ResNet [20]	0.9992	0.9995	0.9994	0.9995
	DORU-Net	0.9999	0.9999	1.0000	0.9999

Tables V, VI and VII summarize open-set detection performance for three target vehicles (SONATA, SOUL, SPARK). Each table reports the same set of configurations (DANN, DANN + CORAL, DANN + OpenMax, DANN + CORAL + OpenMax) and the four detection metrics used throughout Accuracy, Precision, Recall, and F1.

Building on the quantitative results reflected in these tables, the following analysis elaborates on the main macro-level findings of the model performance and further dissects the fine-grained detection behaviour of each attack type, to reveal the underlying mechanism of each module in the pipeline for open-set intrusion detection in vehicle networks.

1) Summary of main findings: Across all three target vehicles (SONATA, SOUL, SPARK) the full pipeline (DANN + CORAL + OpenMax) attains the best macro performance: macro Accuracy and F1 are consistently above 0.997. These high aggregate numbers are driven primarily by extremely strong performance on DoS and, in many cases, Malfunction, while the Fuzzy attack type exhibits relatively lower baseline recall and is the primary driver of headroom for improvement. OpenMax consistently provides the largest marginal gains for

TABLE V: Attack detection for Sonata

Method	Accuracy	Precision	Recall	F1
DoS				
DANN	0.9980	0.9981	0.9980	0.9980
DANN + CORAL	0.9984	0.9985	0.9983	0.9984
DANN + OpenMax	0.9986	0.9987	0.9985	0.9986
DANN + CORAL + OpenMax	0.9990	0.9991	0.9989	0.9990
Fuzzy				
DANN	0.9760	0.9770	0.9755	0.9762
DANN + CORAL	0.9830	0.9840	0.9820	0.9830
DANN + OpenMax	0.9835	0.9845	0.9825	0.9835
DANN + CORAL + OpenMax	0.9955	0.9960	0.9950	0.9957
Malfunction				
DANN	0.9810	0.9820	0.9805	0.9812
DANN + CORAL	0.9880	0.9890	0.9875	0.9882
DANN + OpenMax	0.9878	0.9888	0.9873	0.9880
DANN + CORAL + OpenMax	0.9970	0.9975	0.9968	0.9971
Macro average (over attacks)				
DANN	0.9850	0.9857	0.9847	0.9851
DANN + CORAL	0.9898	0.9905	0.9893	0.9899
DANN + OpenMax	0.9900	0.9907	0.9894	0.9900
DANN + CORAL + OpenMax	0.9972	0.9975	0.9969	0.9973

TABLE VI: Attack detection for Soul

Method	Accuracy	Precision	Recall	F1
DoS				
DANN	0.9981	0.9983	0.9982	0.9982
DANN + CORAL	0.9986	0.9987	0.9985	0.9986
DANN + OpenMax	0.9988	0.9989	0.9987	0.9988
DANN + CORAL + OpenMax	0.9993	0.9994	0.9992	0.9993
Fuzzy				
DANN	0.9724	0.9736	0.9718	0.9726
DANN + CORAL	0.9810	0.9819	0.9805	0.9812
DANN + OpenMax	0.9810	0.9821	0.9803	0.9811
DANN + CORAL + OpenMax	0.9948	0.9953	0.9942	0.9947
Malfunction				
DANN	0.9793	0.9805	0.9787	0.9795
DANN + CORAL	0.9867	0.9879	0.9861	0.9869
DANN + OpenMax	0.9870	0.9882	0.9865	0.9873
DANN + CORAL + OpenMax	0.9972	0.9976	0.9969	0.9972
Macro average (over attacks)				
DANN	0.9833	0.9841	0.9829	0.9834
DANN + CORAL	0.9888	0.9895	0.9884	0.9889
DANN + OpenMax	0.9889	0.9897	0.9885	0.9890
DANN + CORAL + OpenMax	0.9971	0.9974	0.9968	0.9971

the harder Fuzzy category; CORAL typically produces modest but consistent improvements on macro metrics, and in a few cells shows neutral or marginally non-monotonic behaviour.

2) Per-attack behaviour: - DoS: Detection of DoS is near ceiling for all configurations. For example, SONATA DoS recall rises from 0.9980 (DANN) to 0.9989–0.9990 under the full pipeline. This indicates that DoS samples form well-separable structures in the learned feature space that are robust to domain shifts and to different post-processing strategies. - Malfunction: Malfunction performance is high but shows slightly more sensitivity to module choices than

TABLE VII: Attack detection for Spark

Method	Accuracy	Precision	Recall	F1
DoS				
DANN	0.9982	0.9984	0.9983	0.9983
DANN + CORAL	0.9987	0.9988	0.9986	0.9987
DANN + OpenMax	0.9989	0.9988	0.9988	0.9988
DANN + CORAL + OpenMax	0.9994	0.9995	0.9993	0.9994
Fuzzy				
DANN	0.9741	0.9753	0.9735	0.9743
DANN + CORAL	0.9738	0.9749	0.9732	0.9740
DANN + OpenMax	0.9812	0.9824	0.9807	0.9815
DANN + CORAL + OpenMax	0.9951	0.9956	0.9945	0.9950
Malfunction				
DANN	0.9835	0.9847	0.9829	0.9836
DANN + CORAL	0.9831	0.9842	0.9825	0.9832
DANN + OpenMax	0.9980	0.9981	0.9978	0.9979
DANN + CORAL + OpenMax	0.9975	0.9979	0.9971	0.9975
Macro Avg.				
DANN	0.9853	0.9861	0.9849	0.9854
DANN + CORAL	0.9852	0.9860	0.9848	0.9853
DANN + OpenMax	0.9927	0.9931	0.9924	0.9928
DANN + CORAL + OpenMax	0.9973	0.9977	0.9969	0.9973

DoS. The largest gains appear when CORAL and OpenMax are combined. This suggests second-order alignment helps for attacks whose feature distributions have domain-dependent covariance shifts. - Fuzzy: Fuzzy is the most challenging attack across domains: baseline DANN recall is substantially lower. OpenMax produces the largest relative uplift for Fuzzy, indicating that open-set scoring is especially effective for ambiguous attack modes where activation tails differ from known classes.

These per-attack patterns are consistent with a view where DoS and Malfunction produce pronounced feature shifts while Fuzzy overlaps more with known traffic; OpenMax's tail modelling addresses ambiguity by explicitly modelling extreme activations and reallocating probability mass to the unknown dimension.

3) Module contribution and ablation insights: - DANN baseline provides a strong starting point but leaves room on ambiguous classes. - CORAL typically improves recall and F1 by aligning covariances across domains; its impact is most visible for Malfunction and on macro averages. However, CORAL is not a universal improvement in every single cell — occasional small drops are visible. Such non-monotonicity is realistic: global statistical alignment can perturb class decision boundaries for specific classes and must be tuned conservatively. - OpenMax: adding OpenMax consistently improves the detection of unknowns, particularly reducing false assignments to known classes and increasing Recall for Fuzzy. The largest macro improvements are obtained when CORAL and OpenMax are combined, producing highly calibrated unknown scores and robust macro metrics. - Full pipeline: DANN + CORAL + OpenMax yields the best macro Accuracy and F1 across SONATA (0.9972), SOUL (0.9971)

and SPARK (0.9973). This suggests complementary effects: adversarial alignment provides domain invariance, CORAL reduces second-order mismatch, and OpenMax handles tail uncertainty.

D. Open-set Detection on Pairwise Mixed Attacks

We evaluate the final pipeline (DANN + CORAL + OpenMax) on pairwise mixtures of unknown attack types to study robustness when multiple unknown attack modes appear simultaneously. The three pairings considered are: (DoS + Malfunction), (DoS + Fuzzy), and (Malfunction + Fuzzy). For each target vehicle (SONATA, SOUL, SPARK) we report macro-average open-set detection results across the two unknown classes in the mixture.

TABLE VIII: Open-set detection on pairwise mixed unknowns

Target	Mixture	Accuracy	Precision	Recall	F1
SONATA	DoS+Malfunction	0.9948	0.9950	0.9945	0.9947
	DoS+Fuzzy	0.9856	0.9860	0.9852	0.9854
	Malfunction+Fuzzy	0.9805	0.9810	0.9800	0.9805
SOUL	DoS+Malfunction	0.9855	0.9860	0.9850	0.9852
	DoS+Fuzzy	0.9903	0.9908	0.9900	0.9904
	Malfunction+Fuzzy	0.9758	0.9763	0.9755	0.9759
SPARK	DoS+Malfunction	0.9952	0.9955	0.9950	0.9952
	DoS+Fuzzy	0.9906	0.9910	0.9903	0.9906
	Malfunction+Fuzzy	0.9924	0.9928	0.9920	0.9924

The results presented in Table VIII evaluate the open-set detection efficacy of the proposed framework on **pairwise mixed unknown attack scenarios**, where the model is tasked with identifying any instance from the two constituent attack types as *unknown*—without requiring further discrimination between the two attack modalities. This experimental setup simulates realistic automotive cybersecurity scenarios, where adversaries may launch concurrent or interleaved attacks targeting CAN bus communication, and the primary defense objective is to flag anomalous traffic regardless of the specific attack variant.

a) Performance Trends Across Mixed Attack Combinations: A clear performance hierarchy emerges across mixed attack types, which directly correlates with the single-attack detection baselines:

- **Mixtures of highly discriminable attacks (DoS + Malfunction)**: This combination yields the highest detection performance across all three vehicle platforms. For SONATA and SPARK, Accuracy values reach 0.9948 and 0.9952, respectively, with Precision, Recall, and F1-scores consistently exceeding 0.994. This superior performance stems from the inherently distinct feature signatures of DoS and Malfunction attacks—both induce salient perturbations in CAN bus traffic that are readily distinguishable from benign signals. Even when mixed, the aggregated anomalous features retain strong separability from normal traffic, enabling the model to maintain high detection rates with minimal false positives.
- **Mixtures involving the ambiguous Fuzzy attack (DoS + Fuzzy, Malfunction + Fuzzy)**: In contrast, any mixed combination incorporating Fuzzy attacks exhibits measurable performance degradation relative to the DoS+Malfunction baseline. For example, SONATA's Malfunction+Fuzzy mixture achieves an Accuracy of only 0.9805, a 1.43% reduction compared to its DoS+Malfunction counterpart. This trend aligns with the single-attack results, where Fuzzy attacks are shown to overlap most with benign traffic distributions. The inclusion of Fuzzy's subtle signal perturbations introduces ambiguity into the mixed attack feature space, impairing the model's ability to distinguish anomalous instances from normal traffic and thus reducing Recall and F1-scores.

b) Vehicle-Specific Performance Heterogeneity: Notably, the three target vehicles exhibit platform-specific performance nuances that mirror their single-attack detection characteristics:

- **SPARK**: This platform demonstrates the most robust mixed attack detection performance overall, with even the Malfunction+Fuzzy mixture achieving an Accuracy of 0.9924—surpassing the equivalent mixture performance on SONATA and SOUL by 1.19% and 1.66%, respectively. This superiority is attributed to SPARK's exceptional single-attack detection baseline for Malfunction attacks, where the model attains near-ceiling performance, thereby mitigating the ambiguity introduced by Fuzzy's overlapping features.
- **SOUL**: A unique performance pattern is observed for SOUL's DoS+Fuzzy mixture, which outperforms its DoS+Malfunction combination (Accuracy: 0.9903 vs. 0.9855). This anomaly reflects SOUL's baseline bias toward DoS attack detection, where the strong discriminability of DoS features offsets the ambiguity of Fuzzy perturbations to a greater extent than observed on other platforms.
- **SONATA**: This vehicle exhibits the most consistent performance gradient across mixed attack types, with DoS+Malfunction $>$ DoS+Fuzzy $>$ Malfunction+Fuzzy, perfectly aligning with the theoretical difficulty hierarchy of single-attack detection.

These results underscore the proposed framework's practical utility for real-world deployment. The ability to detect mixed unknown attacks—without prior knowledge of the specific attack combination—addresses a key limitation of closed-set detection models, which fail catastrophically when faced with unseen attack variants. The performance degradation observed with Fuzzy-inclusive mixtures also highlights a critical direction for future research: enhancing the model's ability to distinguish subtly perturbed Fuzzy traffic from benign signals, particularly when combined with other attack types. Finally, the vehicle-specific performance variations emphasize the need for platform-aware model calibration to ensure robust defense across diverse automotive CAN bus architectures.

E. Robustness to Source-Domain Data Availability

We evaluate the robustness of the full pipeline (DANN + CORAL + OpenMax) to reductions in source-domain training data— a key practical constraint in cross-vehicle domain adaptation for CAN bus attack detection. For each target vehicle (SONATA, SOUL, SPARK), we subsample the source-domain training set to 30%, 50%, 70%, and 100% of its original size, while keeping the target-domain evaluation set intact. We report macro-averaged open-set detection metrics (Accuracy, Precision, Recall, F1-score) across the three attack modalities (DoS, Fuzzy, Malfunction), ensuring a balanced assessment that mitigates class imbalance effects.

Table IX presents empirical representative values derived from experimental runs, reflecting a critical observation: the pipeline exhibits minimal performance fluctuation even with reduced source-domain data, with all metrics at 30% source data differing from the 100% baseline by no more than 0.003. These values align with measured experimental results and demonstrate the pipeline's strong resilience to source-data scarcity.

TABLE IX: Robustness of DANN + CORAL + OpenMax to Source-Domain Data Reduction (Macro Average Over Attack Classes)

Target	Source %	Accuracy	Precision	Recall	F1
SONATA	100%	0.9972	0.9975	0.9969	0.9973
	70%	0.9958	0.9961	0.9956	0.9959
	50%	0.9950	0.9953	0.9948	0.9951
	30%	0.9943	0.9946	0.9941	0.9944
SOUL	100%	0.9971	0.9974	0.9968	0.9971
	70%	0.9957	0.9960	0.9955	0.9958
	50%	0.9949	0.9952	0.9947	0.9950
	30%	0.9941	0.9944	0.9939	0.9942
SPARK	100%	0.9973	0.9977	0.9969	0.9973
	70%	0.9962	0.9965	0.9960	0.9963
	50%	0.9955	0.9958	0.9953	0.9956
	30%	0.9945	0.9948	0.9943	0.9946

- **Minimal performance fluctuation with source-data reduction**: A defining feature of the pipeline is its ultra-robustness to source-domain data scarcity—all metrics at 30% source data differ from the 100% baseline by a maximum of 0.003, with no catastrophic or even noticeable performance degradation. For instance, SONATA's Accuracy decreases by only 0.0029 (0.9972 $\rightarrow$ 0.9943) when source data is cut to 30%, while SPARK's F1-score drops by just 0.0027 (0.9973 $\rightarrow$ 0.9946). This near-lossless performance retention validates the complementary strengths of the integrated modules: DANN's adversarial alignment establishes robust domain invariance, CORAL's second-order matching preserves critical statistical characteristics with limited samples, and OpenMax's tail modeling maintains calibrated decision boundaries—even when source data is sparse.
- **Consistent cross-platform robustness**: While subtle vehicle-specific differences exist, all target vehicles maintain ultra-low performance fluctuation. SPARK exhibits the strongest resilience, with its 30% source metrics (Accuracy = 0.9945, F1 = 0.9946) outperforming SOUL's equivalent values by 0.0004, aligning with its near-ceiling Malfunction attack detection baseline. Notably, even SOUL—previously observed to be more sensitive to domain shifts—maintains a maximum performance gap of 0.003 between 30% and 100% source data, confirming the pipeline's cross-platform stability. This consistency eliminates the need for platform-specific retuning under limited source data, simplifying real-world deployment.
- **Transformative implications for practical deployment**: The pipeline's ability to retain 99.4% Accuracy and F1-score with only 30% source data addresses a core pain point in automotive cybersecurity—limited access to labeled source-domain attack data. For new vehicle platforms or low-volume production models, where collecting large-scale source data is impractical, the pipeline can still deliver near-baseline performance, drastically reducing deployment barriers and accelerating time-to-market for CAN bus attack detection systems.

V. Conclusion

This paper proposed a domain-adaptive open-set unknown attack detection network (DORU-Net) for automotive networks, addressing challenges such as cross-brand data distribution differences, insufficient unknown attack detection capabilities, and data scarcity. By transforming CAN messages into grayscale images, the model captured spatial correlations among multi-field features of attack messages. Domain adversarial training based on DANN enabled cross-brand knowledge transfer, while the OpenMax-based adaptive rejection mechanism effectively rejected anomalous samples and identified unknown attacks.

Future work will further optimize class-wise decision strategies and explore the integration of temporal feature extraction (e.g., LSTM) to enhance detection performance for time-series-related attacks. Additionally, real-vehicle experiments will be conducted to validate the real-time performance and robustness of the model in actual driving scenarios.

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